

Any time there is a problem to be solved or a decision to be made, our brains must figure out which resources to deploy to meet the challenge. It stands to reason that the brain would like to use tried-and-tested [approaches](#) as a first pass. When you reach for a mug of coffee on the table, you use a normal grip that is consistent with the shape of a typical mug. If this simple approach works, you've freed up the rest of your brain to focus on other tasks (such as talking to a friend). The cerebellum, which is responsible for executing well-learned actions such as this one, is absolutely [doused](#) in serotonergic inputs that [increase](#) its ability to provide rapid, precise feedback to the cortex. In the cortex, serotonin also acts to inhibit cortical output via inhibitory 5-HT₁ receptors – effectively signalling that the cerebellum can digest the information without any cortical help.

Through these processes, serotonin helps the brain continue with an automatic or habitual approach to a situation when that seems to be working well. This is analogous to digestion: if the food you eat can be broken down easily, serotonin will help to ease its passage through the typical digestive process.

Sometimes, however, the in-built approach fails. Maybe the mug is oddly shaped, or turned away from the easy reach of your grasp. Or, going back to the driving example, your usual route home is closed off. In these instances, excitatory 5-HT_{2A} receptors in the cortex are poised to take over. These receptors – which are activated by serotonin – coat excitatory pyramidal cells in the cortex and boost context-specific inputs to the cells. In this way, serotonin increases the range of different cortical cells that can be used to solve the problem at hand – eg, to figure out a way to grasp the mug or to make your way home. This idea maps on to the situation in the gut. If there is a blockage (perhaps caused by overindulgence at a fondue party), then the gut can turn up the peristalsis to deal with it. We envision an analogous situation with cognitive problems: if there is a blockage (eg, your initial approach doesn't work), then you need to create new options for solving the problem.

Why should serotonin play this role in cognitive processing, and not some other neurotransmitter? The answer, like many in evolutionary neuroscience, is difficult to pin down, given how much time has passed since the events that catalysed this putative function. There's nothing particularly special about serotonin compared with other neuromodulatory chemicals, such as dopamine and noradrenaline. They each arise as metabolites of food products and use similar types of cellular mechanisms to enact change in the nervous system. Simply put, they were likely the kinds of fodder that were around when animals (or perhaps single cells) needed a means for conveying a simple message (eg, *We just ate, it's time to digest*) to a complex system of interconnecting parts. What was striking to my colleagues and me, however, was

just how cleanly the idea of digestion and satiation mapped on to the language of cognition, and how fitting the new framing seemed to be for thinking about what serotonin might do (and not do) when the system is pushed to its natural extremes.

This perspective could help to make sense of some other effects of serotonin. For instance, the active component of many psychedelic drugs is the [activation](#) of the excitatory 5-HT_{2A} receptor, albeit in ways that alter the typical timescales upon which the receptors normally act. The effect could be a bit like the brain acting as if previously learned solutions to problems aren't feasible, and doing so in a manner that is mismatched to the current situation – you could be sitting quite comfortably on your couch listening to music, but experience altered visual perceptions induced by the sounds you hear, causing you to process the song in a completely new way. This viewpoint is consistent with a range of recent [imaging](#) and [modelling](#) projects showing that psychedelic agents heighten coordinated activity around the brain, helping to explain the integration of sensory experiences across different modalities.